AI-Enabled Coconut Land Utilization, WCLWD Severity Assessment, Germplasm Identification and Community Intelligence Advisory System.

IT4010 Research Project – 2026



R26-IT-019

Comments for the Progress presentation -01

IT Number	Comment	Action
IT22904164	There is a need to clearly explain and expand the current problem statement and further improve the technical depth of the components involved in the project. Try to add more data to the dataset and improve the model so that it can generate more accurate and reliable results.	improved the problem statement and strengthened the technical implementation by refining the YOLO model, enhancing the dataset preparation, and adding vegetation masking and free-space analysis to estimate available land and approximate additional planting capacity.
IT22236500	Try using different models and evaluate whether better accuracy can be achieved compared to the existing model. The dataset should also be increased to improve the reliability of the results. In addition, minor interface improvements are required, such as automatic palm ID generation and geometric location tracking.	Improved the problem statement and strengthened the technical implementation by expanding the dataset, evaluating different models for better accuracy, and adding automatic palm ID generation and geographic location tracking.
IT22549204	Try using different models and evaluate whether better accuracy can be achieved compared to the existing model. The dataset should also be increased to improve the reliability of the results. .	Balanced the dataset using augmentation and improved the training data Achieved high test accuracy
IT22088314	Try using different models and evaluate whether better accuracy can be achieved compared to the existing model. The dataset should also be increased to improve the reliability of the results.	• Improved the model accuracy while implementing techniques to reduce overfitting and improve reliability.

The Problem





Impact of the Problem on Users and Society

Impact on Farmers

- Late disease detection reduces crop yield and income.
- Dependence on experts slows treatment decisions.

Impact on Agricultural Planning & Disease Monitoring

- Lack of accurate GIS-based land analysis.
- Delayed detection and difficulty identifying diseases like WCLWD.

Impact on Research and Gene Bank Management

- Manual germplasm identification and fragmented records.
- Inefficient data management and slow research processes.

Impact on Community & Economy

- No unified platform for crowdsourced disease reports.
- Reduced coconut production impacts the national economy.

Users & Stakeholders

Coconut Farmers

CRI Authorities

Plantation Managers

Field Officers

System Administrators

Agricultural Researchers



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1. COMMERCIALIZATION PLAN

Scalable AI Platform
Cloud-based, modular system with API access

Partnerships
Collaborate with CRI, govt. agencies, and agri organizations

Go-to-Market
Web & mobile app, workshops, field demos, and agri events

Expansion
Start with coconut sector, expand to other crops

2. TARGET STAKEHOLDERS

- Coconut Research Institute (CRI)
- Plantation Owners & Estate Managers
- Farmers
- Agricultural Departments & NGOs
- Agri Input Suppliers & Service Providers

3. HOW WE SELL

SaaS Model
Web & Mobile Access

PLAN	FREE PLAN (Basic)	SUBSCRIPTION PLAN (Premium)
Features	Basic disease detection	Severity (ABCD & DSI), Localization, Reports, Analytics
Users	Farmers, General Public	Institutions, Plantations, Govt., NGOs
Pricing	Free	Subscription (Monthly / Annual)

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3. ESTIMATED EXPENSES (LKR)

CATEGORY	YEAR 1 (LKR)	YEAR 2 (LKR)	YEAR 3 (LKR)
AI Model Development & Training	600,000	300,000	300,000
Cloud Infrastructure & Hosting	400,000	600,000	700,000
Data Collection & Annotation	250,000	250,000	250,000
Software Development (Web & Mobile)	500,000	300,000	300,000
Marketing & Outreach	200,000	250,000	300,000
Partnerships & Training	150,000	200,000	250,000
Operational Costs	200,000	250,000	300,000
TOTAL (LKR)	**2,300,000**	**2,150,000**	**2,400,000**

** Expenses can vary based on scale and partnerships.*

4. RETURN & REVENUE MODEL

REVENUE STREAMS

- Subscription plans (monthly / annual)
- Institutional licensing
- Data analytics & reporting services
- Custom AI model & integration services
- Training & consultancy

RETURN STRATEGY

- Break-even expected in Year 2
- Reinvest in model improvement, data expansion & partnerships
- Scale to other crops and regions

PROFITABILITY OUTLOOK

Growing subscription base and data services drive recurring revenue and long-term sustainability.

5. PRICING & SUBSCRIPTION PLANS

FEATURES	FREE PLAN (Basic)	SUBSCRIPTION PLAN (Premium)
Disease Detection	✓ Basic detection	✓ Advanced detection & severity (ABCD & DSI)
Disease Localization	✗ Not included	✓ WSSS localization
Reports & Analytics	✗ Not included	✓ Detailed reports & analytics
GenBank Access	✗ Limited	✓ Full GenBank data access
Data Download	✗ Not included	✓ Download reports & insights
Support	✓ Community support	✓ Priority support
PRICE	**LKR 0 (Free)**	**LKR 1,500 / month or LKR 15,000 / year**

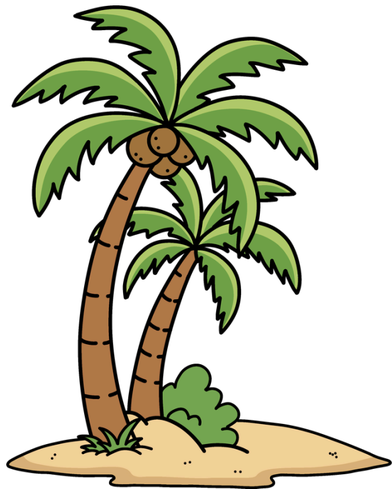
OUR GOAL: To deliver intelligent, data-driven solutions that empower stakeholders, improve productivity, and ensure a sustainable coconut future.

EMPOWER. DETECT. ADVISE.
For a Smarter and Healthier Coconut Future!

AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

GANTT CHART

TASK	OCT	NOV	DEC	JAN	FEB	MAR	MAY	JUN	JUL	AUG	SEP	OCV	DEC
1. Group Registration			20th October 2025										
2. Topic Selection													
3. TAF Documentation													
4. TAF Submission				05th January 2026									
5. Proposal Report					15th March 2026								
6. Proposal Present					16th March 2026								
7. Implementation													
8. Progress Presentation I							4th - 6th May 2026						
9. Research Paper													
10. Testing & Finalize the Project										12th - October 2026			
11. Progress Presentation II										12th - 14th October 2026			
12. Web Assessment											11th October 2026		
13. Final Presentation & Viva													1st December 2026
14. Final Report													



THANK YOU!

